Year 12 Further Maths - Roots of polynomials

Similar questions - Conjugate roots, factors and loci

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

(a) Express $(3+2\mathrm{i})^3$ in the form $a+b\mathrm{i}$. **[3]**

(b) Hence verify that $3+2\mathrm{i}$ is a root of the equation $z^3-8z^2+25z-26=0$. **[3]**

(c) Express $z^3-8z^2+25z-26$ as the product of a linear factor and a quadratic factor with real coefficients. **[4]**

2 In this question you must show detailed reasoning. You are given that $\mathrm{f}(z)=z^4-4z^3+18z^2+20z+125$ and that $3+4\mathrm{i}$ is a root of $\mathrm{f}(z)=0$.

(a) Express $\mathrm{f}(z)$ as the product of two quadratic factors with integer coefficients. **[5]**

(b) Solve $\mathrm{f}(z)=0$. **[3]**

Two loci are $C_1=\{z:|z|=r_1\}$ and $C_2=\{z:|z|=r_2\}$ where $r_1>r_2$, with two of the roots on each. R is the region between C_1 and C_2.

(c) Find the exact area of R. **[4]**

(d) ω is the sum of all the roots of $\mathrm{f}(z)=0$. Determine whether the point representing ω lies in R. **[2]**

3 In this question you must show detailed reasoning. The equation $x^4-6x^3-14x^2+114x-455=0$ has a root $2-3\mathrm{i}$. Find the other three roots. **[6]**

4 In this question you must show detailed reasoning.

(a) Use an algebraic method to find the square roots of $-5+12\mathrm{i}$. **[5]**

(b) By finding the cube of one of your answers to part (a), determine a cube root of $\dfrac{-46+9\mathrm{i}}{8}$. Give your answer in the form $a+b\mathrm{i}$. **[3]**

Question 1
(a) $(3+2\mathrm{i})^3 = 27+54\mathrm{i}-36-8\mathrm{i} = -9+46\mathrm{i}$
(b) $(3+2\mathrm{i})^2 = 5+12\mathrm{i}$. $-9+46\mathrm{i}-8(5+12\mathrm{i})+25(3+2\mathrm{i})-26 = (-9-40+75-26)+(46-96+50)\mathrm{i} = 0$
(c) The conjugate $3-2\mathrm{i}$ is also a root, giving the quadratic factor $z^2-6z+13$. Then $z^3-8z^2+25z-26 = (z-2)(z^2-6z+13)$

Question 2
(a) $3-4\mathrm{i}$ is also a root: quadratic factor $z^2-6z+25$. Dividing: $\mathrm{f}(z) = (z^2-6z+25)(z^2+2z+5)$
(b) $z = 3\pm 4\mathrm{i}$ and $z^2+2z+5=0 \Rightarrow z = -1\pm 2\mathrm{i}$
(c) $|3\pm 4\mathrm{i}| = 5$ and $|-1\pm 2\mathrm{i}| = \sqrt{5}$. Area $= \pi(25-5) = 20\pi$
(d) Sum of roots $= 4$ (from the z^3 coefficient), $|\omega| = 4$ and $\sqrt{5} < 4 < 5$, so ω lies in R

Question 3
Conjugate $2+3\mathrm{i}$ is also a root: factor $(x^2-4x+13)$. Dividing: $x^4-6x^3-14x^2+114x-455 = (x^2-4x+13)(x^2-2x-35) = (x^2-4x+13)(x-7)(x+5)$. Other roots: $2+3\mathrm{i}$, 7, -5

Question 4
(a) $(x+y\mathrm{i})^2 = -5+12\mathrm{i} \Rightarrow x^2-y^2 = -5$, $xy = 6$. $y = \frac{6}{x}$: $x^4+5x^2-36 = 0 \Rightarrow (x^2+9)(x^2-4) = 0$, $x = \pm 2$, $y = \pm 3$. Square roots: $\pm(2+3\mathrm{i})$

(b) $(2+3\mathrm{i})^3 = (2+3\mathrm{i})(-5+12\mathrm{i}) = -10+24\mathrm{i}-15\mathrm{i}-36 = -46+9\mathrm{i}$. Dividing by $8 = 2^3$: $\left(\frac{2+3\mathrm{i}}{2}\right)^3 = \dfrac{-46+9\mathrm{i}}{8}$, so a cube root is $1+\frac{3}{2}\mathrm{i}$